

Q- List three differences between static routing and dynamic routing in a table format, using real-world analogies from apps like Google Maps or Swiggy to make each point relatable.

| Point                          | Static Routing                                                                                                                                                                                                                  | Dynamic Routing                                                                                                                                                                                                                                          |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Route Selection             | Routes are <b>manually configured</b> by the network administrator. <b>Analogy:</b> Like entering a fixed route in <b>Google Maps</b> yourself—you always follow the selected road unless you change it.                        | Routes are <b>automatically learned and updated</b> by routing protocols. <b>Analogy:</b> Google Maps automatically suggests a new route when there is a traffic jam.                                                                                    |
| 2. Handling Changes            | Does <b>not automatically react</b> to network failures or changes. <b>Analogy:</b> In <b>Swiggy</b> , if a delivery road is blocked, a rider following a fixed route would need someone to manually give a new route.          | Automatically finds an <b>alternative route</b> when a network path fails. <b>Analogy:</b> Swiggy can redirect a delivery rider when the usual road becomes unavailable.                                                                                 |
| 3. Configuration & Scalability | Easy to configure in a <b>small network</b> , but becomes difficult to manage as the network grows. <b>Analogy:</b> Planning routes manually for a few Swiggy deliveries is easy, but managing thousands manually is difficult. | Requires more initial configuration but is better for <b>large networks</b> because routers exchange routing information automatically. <b>Analogy:</b> Google Maps manages routes for millions of users without each user manually planning every road. |